Digit-Transfer Separation, Macroscopic Defect Laws, and a Determinant-One Gap Ladder

New progress toward the Alaoglu–Erdős conjecture

OpenAI GPT-5.6 Pro

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Status declaration. This report does *not* claim a proof of the Alaoglu–Erdős conjecture. It develops a new, self-contained digit-transfer framework, proves several rigorous separation and defect theorems, constructs an exact synchronized continued-fraction ladder, and isolates two sharply formulated missing arithmetic inputs. Statements labeled as theorems or propositions below are proved; proposed closing steps are explicitly labeled as problems or criteria.

Abstract

Let

$$\vartheta = \log_2 3, \qquad \varrho = \vartheta^{-1} = \log_3 2.$$

The Alaoglu–Erdős conjecture asks whether integers $M = 2^x$ and $A = 3^x$ can occur for a nonintegral real x. The supplied unified report already develops the standard transcendence barrier, primitive normalization, pure-radical reduction, determinant and Padé constructions, local methods, and numerous approximation-theoretic routes. The present report deliberately avoids reproducing that material.

Our first new device is the exact binary-to-ternary digit-transfer map

$$\Phi\Big(\sum \epsilon_j 2^j\Big) = \sum \epsilon_j 3^j$$

and its ternary-to-binary retraction

$$\Psi\Big(\sum \delta_j 3^j\Big) = \sum \delta_j 2^j.$$

We prove their carry calculus, sharp global power bounds, summatory and Dirichlet-series identities, and a two-prime-adic retraction theorem. For a hypothetical pair $A = M^\vartheta$, the defects

$$D_k = A^k - \Phi(M^k), \qquad E_k = \Psi(A^k) - M^k$$

are positive integers. They satisfy exact interval-valuation identities, $0 < E_k \le D_k$, and $E_k \le 2D_k^\varrho$.

The strongest structural result is a *prefix-separation theorem*: when the base-2 expansion of M^k and the base-3 expansion of A^k are aligned at their common most-significant position, their common prefix is either empty or exactly a leading 1 followed by zeros. At the first mismatch, the ternary digit is strictly larger. Combining this with an explicit lower bound for a linear form in $\log M$ and $\log 2$ places the first mismatch within $O(\log k)$ places of the leading digit. We also prove positive-density and effective all-k macroscopic lower bounds for both defects.

A second new construction starts from the pure-radical normalization in the supplied report. Continued-fraction convergents to a single logarithm produce simultaneous near-power gaps at bases 2 and 3 with the same small logarithmic error and adjacent exponent determinant exactly one. On one sign subsequence, digit transfer gives an exact "gap-descent square". The square is not contractive: we quantify the scale mismatch that prevents it from closing the conjecture. This no-go calculation leaves a precise research target—a prime-transfer or carry-rigidity theorem strong enough to connect the two synchronized gaps.

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1 Scope, imported facts, and nonduplication

1.1 The conjecture and hypothetical notation

The conjecture is

$$x \in \mathbb{R}, \quad 2^x \in \mathbb{Z}, \quad 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}. \tag{1}$$

Write

$$M = 2^x, \qquad A = 3^x, \qquad \vartheta = \log_2 3, \qquad \varrho = \log_3 2. \tag{2}$$

Then

$$A = M^\vartheta, \qquad M = A^\varrho, \qquad \log_2 M = \log_3 A = x. \tag{3}$$

A nonintegral rational x is impossible by unique factorization, so a hypothetical counterexample has irrational x. Standard transcendence theorems moreover force it beyond the algebraic-irrational case. These reductions, and their relationship to the four-exponentials frontier, are part of the supplied report and are not re-proved here.

Throughout the main digit-transfer analysis we therefore make the following standing hypothesis solely for contradiction-seeking purposes.

Definition 1.1 (Hypothetical pair)**.** A *hypothetical pair* is a pair of integers $M, A > 1$ such that $A = M^\vartheta$, with neither M a power of 2 nor A a power of 3. We set

$$\beta = \log_2 M = \log_3 A \notin \mathbb{Q}.$$

1.2 What is imported from the supplied report

Only the following established reductions are imported.

1. A counterexample, if one exists, may be organized through a least positive generator and a primitive normalization.
2. In one convenient branch there are integers $a, d \geq 0$, an odd integer $w > 1$, and an integer $A > 1$ such that
$$M = 2^a w^d, \qquad \eta = w^\vartheta, \qquad \eta^d = \frac{A}{3^a}, \tag{4}$$
and η has exact algebraic degree d.
3. Generic adjacent near-power gaps admit a determinant-controlled gcd bound: after removing primes dividing the two bases, the gcd of two gaps is controlled by the determinant of their exponent vectors. We invoke this only in section 10, where our new exponent vectors have determinant exactly one.

No determinant expansions, Padé approximants, finite-difference constructions, automatic-sequence surveys, or local analytic arguments from the supplied manuscript are repeated. The maps Φ and Ψ, the prefix-separation theorem, the defect interval identities, and the synchronized determinant-one ladder below are new organizational devices relative to that manuscript.

1.3 Why digit transfer is a natural test

The equality $A^k = (M^k)^\vartheta$ converts each binary atom exactly into a ternary atom:

$$(2^j)^\vartheta = 3^j.$$

For a binary expansion $M^k = \sum \epsilon_j 2^j$, convexity gives

$$A^k = \left(\sum \epsilon_j 2^j\right)^\vartheta > \sum \epsilon_j 3^j$$

unless there is only one nonzero bit. The right side is an integer with ternary digits $0, 1$. Conversely, concavity of t^{ϱ} compares the ternary expansion of A^k with M^k. This creates two integral defects whose arithmetic can be studied exactly, rather than only estimated by real analysis.

2 The two digit-transfer maps

Definition 2.1 (Finite digit-transfer maps)**.** For $n \in \mathbb{N}_0$, write its canonical binary and ternary expansions as

$$n = \sum_{j\geq 0} \epsilon_j(n) 2^j, \quad \epsilon_j(n) \in \{0,1\}, \qquad n = \sum_{j\geq 0} \delta_j(n) 3^j, \quad \delta_j(n) \in \{0,1,2\}.$$

Define

$$\Phi(n) = \sum_{j\geq 0} \epsilon_j(n) 3^j, \qquad \Psi(n) = \sum_{j\geq 0} \delta_j(n) 2^j. \tag{5}$$

Thus Φ reinterprets a binary word as a ternary word, while Ψ reinterprets a ternary word as a binary digit string that is then evaluated without first normalizing digits 2.

2.1 Radix recurrences

Proposition 2.2 (Exact recurrences)**.** *For all $n \geq 0$,*

$$\Phi(2n) = 3\Phi(n), \qquad \Phi(2n+1) = 3\Phi(n) + 1, \tag{6}$$

$$\Psi(3n+r) = 2\Psi(n) + r \qquad (r = 0, 1, 2). \tag{7}$$

Proof. Appending a binary digit $r \in \{0, 1\}$ multiplies the transferred ternary word by 3 and adds r. Appending a ternary digit $r \in \{0, 1, 2\}$ multiplies its binary evaluation by 2 and adds r. □

The increment sequences are unexpectedly simple.

Theorem 2.3 (Valuation increments)**.** *For every integer $n \geq 1$,*

$$\Phi(n) - \Phi(n-1) = \frac{3^{v_2(n)} + 1}{2}, \tag{8}$$

$$\Psi(n) - \Psi(n-1) = 2 - 2^{v_3(n)}. \tag{9}$$

Proof. Let $r = v_2(n)$. Passing from $n-1$ to n changes the trailing binary word $0\,1^r$ to $1\,0^r$. Under Φ, the increment is

$$3^r - \sum_{j=0}^{r-1} 3^j = 3^r - \frac{3^r - 1}{2} = \frac{3^r + 1}{2}.$$

For $r = v_3(n)$, the trailing ternary word changes from $0\,2^r$ to $1\,0^r$. Its binary evaluation changes by

$$2^r - 2\sum_{j=0}^{r-1} 2^j = 2^r - 2(2^r - 1) = 2 - 2^r.$$

□

Corollary 2.4 (Summatory valuation identities)**.** *For $N \geq 1$,*

$$\Phi(N) = \frac{N}{2} + \frac{1}{2} \sum_{n\leq N} 3^{v_2(n)}, \tag{10}$$

$$\Psi(N) = 2N - \sum_{n\leq N} 2^{v_3(n)}. \tag{11}$$

Proof. Sum equations (8) and (9) from 1 to N, using $\Phi(0) = \Psi(0) = 0$. □

2.2 Dirichlet series and log-periodic scaling

Proposition 2.5 (Valuation Dirichlet series). *For a prime p, a positive real c, and $\Re(s) > \max\{1, \log_p c\}$,*

$$\sum_{n\geq 1} \frac{c^{v_p(n)}}{n^s} = \zeta(s)\frac{1-p^{-s}}{1-cp^{-s}}. \tag{12}$$

In particular,

$$\sum_{n\geq 1} \frac{3^{v_2(n)}}{n^s} = \zeta(s)\frac{1-2^{-s}}{1-3\,2^{-s}}, \qquad \Re(s) > \vartheta, \tag{13}$$

$$\sum_{n\geq 1} \frac{2^{v_3(n)}}{n^s} = \zeta(s)\frac{1-3^{-s}}{1-2\,3^{-s}}, \qquad \Re(s) > 1. \tag{14}$$

Proof. Write $n = p^j m$ with $p \nmid m$. Then

$$\sum_{n\geq 1} \frac{c^{v_p(n)}}{n^s} = \sum_{j\geq 0}(cp^{-s})^j \sum_{p\nmid m} \frac{1}{m^s} = \frac{1}{1-cp^{-s}}(1-p^{-s})\zeta(s).$$

□

The extra pole lattice of equation (13) on $\Re(s) = \vartheta$ reflects the exact scaling $\Phi(2n) = 3\Phi(n)$. In equation (14), the pole of $\zeta(s)$ at $s = 1$ gives the linear term $2N$ in equation (11); subtracting it reveals the lower-order N^{ϱ}-scale fractal $\Psi(N)$.

2.3 Carry monotonicity

Theorem 2.6 (Carry inequalities). *For all nonnegative integers m, n,*

$$\Phi(m+n) \geq \Phi(m) + \Phi(n), \qquad \Phi(mn) \geq \Phi(m)\Phi(n), \tag{15}$$

$$\Psi(m+n) \leq \Psi(m) + \Psi(n), \qquad \Psi(mn) \leq \Psi(m)\Psi(n). \tag{16}$$

Moreover, Φ is strictly increasing.

Proof. Before carrying, addition or multiplication of binary digit strings produces nonnegative integer coefficients at the weights 2^j. A binary carry replaces $2 \cdot 2^j$ by 2^{j+1}. Under evaluation at 3, the same operation replaces $2 \cdot 3^j$ by 3^{j+1}, increasing the value by 3^j. Normalizing all carries therefore proves the two inequalities for Φ. Strict monotonicity also follows from equation (8).

For ternary strings, a carry replaces $3 \cdot 3^j$ by 3^{j+1}. Under evaluation at 2, it replaces $3 \cdot 2^j$ by 2^{j+1}, decreasing the value by 2^j. Normalizing all ternary carries proves both inequalities for Ψ. □

Warning 2.7. The map Ψ is not order-preserving. Indeed, its increment in equation (9) is negative whenever $v_3(n) \geq 2$. Any argument that applies monotonicity to Ψ is therefore invalid.

2.4 Composition and a two-prime-adic retraction

Theorem 2.8 (Retraction identities). *For every $n \geq 0$,*

$$\Psi(\Phi(n)) = n. \tag{17}$$

For every $a \geq 0$,

$$\Phi(\Psi(a)) \geq a, \tag{18}$$

with equality if and only if the ternary expansion of a uses only the digits 0 and 1. Consequently

$$H = \Phi \circ \Psi \quad \textit{satisfies} \quad H^2 = H. \tag{19}$$

Proof. The ternary digits of $\Phi(n)$ are exactly the binary digits of n, which proves equation (17). To compute $\Phi(\Psi(a))$, first regard the ternary digits $\delta_j(a) \in \{0,1,2\}$ as coefficients at binary weights and then normalize binary carries. Every carry increases evaluation at 3, by the proof of theorem 2.6. Equality occurs precisely when no digit 2 and no subsequent carry is present, namely when all ternary digits lie in $\{0,1\}$. Finally, $H^2 = \Phi\Psi\Phi\Psi = \Phi\Psi = H$. □

The finite maps extend naturally to incompatible local fields.

Theorem 2.9 (Two-prime-adic extension)**.** *The formula*

$$\Phi\left(\sum_{j\geq 0} \epsilon_j 2^j\right) = \sum_{j\geq 0} \epsilon_j 3^j$$

defines a continuous injection $\Phi : \mathbb{Z}_2 \to \mathbb{Z}_3$, whose image is the 3-adic Cantor set

$$\mathcal{C}_3 = \left\{\sum_{j\geq 0} \epsilon_j 3^j : \epsilon_j \in \{0,1\}\right\}.$$

The formula

$$\Psi\left(\sum_{j\geq 0} \delta_j 3^j\right) = \sum_{j\geq 0} \delta_j 2^j$$

defines a continuous surjection $\Psi : \mathbb{Z}_3 \to \mathbb{Z}_2$, and $\Psi \circ \Phi = \mathrm{id}_{\mathbb{Z}_2}$. If $x, y \in \mathbb{Z}_2$, then

$$v_3(\Phi(x) - \Phi(y)) = v_2(x - y). \tag{20}$$

If $a \equiv b \pmod{3^r}$, then

$$\Psi(a) \equiv \Psi(b) \pmod{2^r}. \tag{21}$$

Proof. Both series converge in their target local fields. The first differing binary digit of x, y has coefficient ± 1, so it remains the first nonzero ternary coefficient in the difference; this proves equation (20) and injectivity. Surjectivity and the retraction identity follow by choosing ternary digits in $\{0,1\}$. Congruence equation (21) follows because the first r ternary digits determine the output modulo 2^r. □

3 Sharp archimedean power bounds

The elementary bounds below are optimal in their constants and will be used repeatedly.

Theorem 3.1 (Sharp global bounds)**.** *For every integer $n \geq 1$,*

$$\frac{1}{2} n^{\vartheta} \leq \Phi(n) \leq n^{\vartheta}, \tag{22}$$

and

$$n^{\varrho} \leq \Psi(n) \leq 2n^{\varrho}. \tag{23}$$

Equality in the upper bound for Φ, or in the lower bound for Ψ, occurs exactly when n is a power of the relevant source base (2 for Φ, 3 for Ψ). The constants $1/2$ and 2 are best possible.

Proof. Write $n = \sum \epsilon_j 2^j$. Since $\vartheta > 1$,

$$n^{\vartheta} = \left(\sum \epsilon_j 2^j\right)^{\vartheta} \geq \sum \epsilon_j (2^j)^{\vartheta} = \Phi(n),$$

with equality only when a single bit is nonzero.

For the lower bound, use induction on the binary length. Write $n = 2^L + r$, $0 \le r < 2^L$, and set $t = r/2^L$. Assuming $\Phi(r) \ge r^\vartheta/2$,

$$\Phi(n) = 3^L + \Phi(r) \ge 3^L \left(1 + \frac{1}{2}t^\vartheta\right).$$

The function $2 + t^\vartheta - (1+t)^\vartheta$ is decreasing on $[0,1]$ and vanishes at 1, because $2^\vartheta = 3$. Hence

$$1 + \frac{1}{2}t^\vartheta \ge \frac{1}{2}(1+t)^\vartheta,$$

which proves the induction.

For Ψ, write $n = \sum \delta_j 3^j$. Since $0 < \varrho < 1$, subadditivity gives

$$n^\varrho \le \sum_j (\delta_j 3^j)^\varrho = \sum_j \delta_j^\varrho 2^j \le \sum_j \delta_j 2^j = \Psi(n).$$

Equality requires exactly one digit, equal to 1.

For the upper bound, induct on ternary length. Write $n = d3^L + r$, where $d \in \{1,2\}$, $0 \le r < 3^L$, and $t = r/3^L$. The induction hypothesis gives

$$\Psi(n) = d2^L + \Psi(r) \le 2^L(d + 2t^\varrho).$$

For $d = 1, 2$, the function

$$2(d+t)^\varrho - d - 2t^\varrho$$

is decreasing on $[0,1]$, and its value at 1 is nonnegative (the $d = 2$ value is exactly 0, since $3^\varrho = 2$). Therefore $d + 2t^\varrho \le 2(d+t)^\varrho$, completing the induction.

Finally, at $n = 2^L - 1$, $\Phi(n) = (3^L - 1)/2$, so the ratio in the lower bound tends to $1/2$. At $n = 3^L - 1$, $\Psi(n) = 2(2^L - 1)$, and the ratio in the upper bound tends to 2. □

Corollary 3.2 (Root limits)**.** *For every integer $m \ge 2$,*

$$\lim_{k\to\infty} \Phi(m^k)^{1/k} = m^\vartheta, \tag{24}$$

$$\lim_{k\to\infty} \Psi(m^k)^{1/k} = m^\varrho. \tag{25}$$

Proof. Take k-th roots in equations (22) and (23). □

3.1 Quantitative one-step convexity and concavity

Proposition 3.3 (Binary leading-atom defect)**.** *If $n = 2^L + r$ with $1 \le r < 2^L$, then*

$$n^\vartheta - \Phi(n) \ge 3^L \left[\left(1 + \frac{r}{2^L}\right)^\vartheta - 1 - \left(\frac{r}{2^L}\right)^\vartheta\right] \tag{26}$$

$$\ge (\vartheta - 1)3^L \frac{r}{2^L} \ge (\vartheta - 1)\left(\frac{3}{2}\right)^L. \tag{27}$$

Proof. Use $\Phi(r) \le r^\vartheta$. For $f(t) = (1+t)^\vartheta - 1 - t^\vartheta$, one has

$$f'(t) = \vartheta\big((1+t)^{\vartheta-1} - t^{\vartheta-1}\big) \ge \frac{\vartheta}{2} > \vartheta - 1 \qquad (0 \le t \le 1),$$

where the minimum occurs at $t = 1$ and $2^{\vartheta-1} = 3/2$. Thus $f(t) \ge (\vartheta - 1)t$. □

Proposition 3.4 (Ternary leading-atom defect). *If $n = d3^L + r$, where $d \in \{1,2\}$ and $0 \le r < 3^L$, then*

$$\Psi(n) - n^{\varrho} \ge 2^L \left[d + \left(\frac{r}{3^L}\right)^{\varrho} - \left(d + \frac{r}{3^L}\right)^{\varrho} \right]. \tag{28}$$

For $d = 1$,

$$\Psi(n) - n^{\varrho} \ge (2 - 2^{\varrho}) 2^L \left(\frac{r}{3^L}\right)^{\varrho}. \tag{29}$$

For $d = 2$,

$$\Psi(n) - n^{\varrho} \ge (2 - 2^{\varrho}) 2^L. \tag{30}$$

Proof. Use $\Psi(r) \ge r^{\varrho}$. For $d = 1$, the function

$$g(t) = \frac{(1+t)^{\varrho} - 1}{t^{\varrho}}$$

is increasing on $(0, 1]$, because the sign of its derivative is the sign of $1 - (1+t)^{\varrho - 1} > 0$. Hence $g(t) \le g(1) = 2^{\varrho} - 1$, proving equation (29). For $d = 2$, the expression $2 + t^{\varrho} - (2+t)^{\varrho}$ is increasing in t and at $t = 0$ equals $2 - 2^{\varrho}$. □

4 The integral defect system

Assume henceforth that (M, A) is a hypothetical pair in the sense of definition 1.1. Define, for $k \ge 1$,

$$C_k = \Phi(M^k), \qquad D_k = A^k - C_k, \qquad E_k = \Psi(A^k) - M^k, \qquad K_k = \Phi(\Psi(A^k)) - A^k. \tag{31}$$

Proposition 4.1 (Positivity). *For every $k \ge 1$,*

$$D_k > 0, \qquad E_k > 0, \qquad K_k \ge 0. \tag{32}$$

Proof. The integer M^k is not a power of 2, for otherwise M would be a power of 2. Strictness in the convexity proof of theorem 3.1 gives $\Phi(M^k) < (M^k)^{\vartheta} = A^k$. Similarly, A^k is not a power of 3, so $\Psi(A^k) > (A^k)^{\varrho} = M^k$. The assertion for K_k is equation (18). □

4.1 Exact interval-valuation identities

Theorem 4.2 (Defect interval identities). *For every $k \ge 1$,*

$$E_k = 2D_k - \sum_{C_k < n \le A^k} 2^{v_3(n)}, \tag{33}$$

$$D_k + K_k = \frac{1}{2}\left(E_k + \sum_{M^k < n \le M^k + E_k} 3^{v_2(n)} \right). \tag{34}$$

Consequently,

$$0 < E_k \le D_k \tag{35}$$

and

$$E_k \le \Psi(D_k) \le 2D_k^{\varrho}. \tag{36}$$

Proof. Because $\Psi(C_k) = \Psi(\Phi(M^k)) = M^k$, telescoping equation (9) over $(C_k, A^k]$ gives

$$E_k = \Psi(A^k) - \Psi(C_k) = \sum_{C_k < n \le A^k} (2 - 2^{v_3(n)}),$$

which is equation (33). There are D_k terms in the sum and each $2^{v_3(n)} \ge 1$, proving $E_k \le D_k$.

Next, $\Psi(A^k) = M^k + E_k$, so telescoping equation (8) gives

$$\Phi(\Psi(A^k)) - \Phi(M^k) = \frac{1}{2} \sum_{M^k < n \le M^k + E_k} (3^{v_2(n)} + 1).$$

The left side is $(A^k + K_k) - (A^k - D_k) = D_k + K_k$, proving equation (34).

Finally, subadditivity of Ψ gives

$$M^k + E_k = \Psi(C_k + D_k) \le \Psi(C_k) + \Psi(D_k) = M^k + \Psi(D_k),$$

and equation (23) gives the last inequality. $\square$

Remark 4.3 (What the interval identities say)**.** Equation equation (33) converts the archimedean convexity defect D_k into an exact excess of 3-adic valuation weights on a consecutive interval. Equation equation (34) converts the return defect into a corresponding 2-adic valuation sum. A proof by this route would need more than average-order information: it would have to exploit the special endpoints $\Phi(M^k)$, A^k, M^k, and $\Psi(A^k)$.

4.2 Semigroup inequalities

Proposition 4.4 (Defect semigroup laws)**.** *Set*

$$d_k = \frac{D_k}{A^k}, \qquad e_k = \frac{E_k}{M^k}.$$

Then for all $k, \ell \ge 1$,

$$d_{k+\ell} \le d_k + d_\ell - d_k d_\ell, \tag{37}$$
$$e_{k+\ell} \le e_k + e_\ell + e_k e_\ell. \tag{38}$$

Equivalently, $1 - d_k$ is supermultiplicative and $1 + e_k$ is submultiplicative.

Proof. By equation (15),

$$\Phi(M^{k+\ell}) \ge \Phi(M^k)\Phi(M^\ell).$$

Divide by $A^{k+\ell}$. By equation (16),

$$\Psi(A^{k+\ell}) \le \Psi(A^k)\Psi(A^\ell),$$

and division by $M^{k+\ell}$ gives the second assertion. $\square$

5 The digit-difference polynomial and borrow rigidity

Let

$$M^k = \sum_{j=0}^{L_k} \epsilon_{k,j} 2^j, \qquad A^k = \sum_{j=0}^{L_k} \delta_{k,j} 3^j,$$

where leading zero digits are inserted as needed. The equality of the two upper indices will be proved in section 6. For the moment, take a common sufficiently large upper index.

Definition 5.1 (Difference polynomial)**.** Define

$$P_k(z) = \sum_j (\delta_{k,j} - \epsilon_{k,j}) z^j \in \mathbb{Z}[z]. \tag{39}$$

Proposition 5.2 (Two evaluations of one word). *For every $k \geq 1$,*

$$P_k(3) = D_k, \qquad P_k(2) = E_k. \tag{40}$$

If $J_k = \deg P_k$, then its leading coefficient is positive. More precisely, it lies in $\{1, 2\}$.

Proof. The first identity is the definition of D_k. For the second,

$$P_k(2) = \Psi(A^k) - \Psi(\Phi(M^k)) = \Psi(A^k) - M^k = E_k.$$

If the leading coefficient at degree J were -1, then even the largest possible lower contribution would be

$$-3^J + 2\sum_{j<J} 3^j = -1 < 0,$$

contradicting $D_k > 0$. The coefficient cannot be less than -1, and its positive possibilities are $1, 2$. □

Lemma 5.3 (Leading-coefficient lower bounds). *Let $P(z) = \sum_{j=0}^{J} c_j z^j$ with $c_j \in \{-1, 0, 1, 2\}$ and positive leading coefficient. Then*

$$P(3) \geq \frac{3^J + 1}{2}, \tag{41}$$

$$P(2) \geq 1. \tag{42}$$

If $c_J = 2$, then

$$P(2) \geq 2^J + 1. \tag{43}$$

Proof. The minimum at 3 occurs when all lower coefficients equal -1:

$$3^J - \sum_{j<J} 3^j = \frac{3^J + 1}{2}.$$

At 2, the same calculation gives 1 for leading coefficient 1 and $2^J + 1$ for leading coefficient 2. □

The exceptional way in which $P(2)$ can be small while $P(3)$ is large is completely rigid.

Theorem 5.4 (Borrow-block lemma). *Let $P(z) = \sum_{i=0}^{J} c_i z^i$ be as in lemma 5.3, and set $E = P(2) > 0$.*

If $c_J = 1$, then

$$E = 1 + \sum_{i<J} (c_i + 1)2^i. \tag{44}$$

If $c_J = 2$, then

$$E = 2^J + 1 + \sum_{i<J} (c_i + 1)2^i. \tag{45}$$

Consequently, if $E < 2^s$ with $s \leq J$, then necessarily $c_J = 1$ and

$$c_i = -1 \qquad (s \leq i < J). \tag{46}$$

For the digit polynomial P_k, this means

$$(\delta_{k,i}, \epsilon_{k,i}) = (0, 1) \qquad (s \leq i < J_k). \tag{47}$$

Proof. Use $\sum_{i<J} 2^i = 2^J - 1$ to add and subtract the all-ones lower word. Every coefficient $c_i + 1$ is nonnegative. If the total is below 2^s, all terms at positions $i \geq s$ must vanish. Since $c_i = \delta_{k,i} - \epsilon_{k,i}$, the value -1 occurs only for the pair $(0, 1)$. □

Corollary 5.5 (Congruence form of a long borrow block)**.** *Under the hypotheses of theorem 5.4, if $E_k < 2^s$ and $s \le J_k$, then there are integers u, v with*

$$1 \le u \le 2^s, \qquad 0 \le v < 3^s, \tag{48}$$

such that

$$M^k \equiv -u \pmod{2^{J_k}}, \qquad A^k \equiv v \pmod{3^{J_k}}. \tag{49}$$

Proof. The binary bits in positions $s, \ldots, J_k - 1$ are all 1, so modulo 2^{J_k} the lower word is $2^{J_k} - u$ with $1 \le u \le 2^s$. The ternary digits in the same positions vanish, leaving a residue below 3^s. □

5.1 The exact carry mask for the return defect

Theorem 5.6 (Carry-mask formula)**.** *Let $a = \sum_{j\ge0} \delta_j 3^j$, and normalize the digit string $\delta_j \in \{0, 1, 2\}$ in base 2 by carry bits $c_j \in \{0, 1\}$:*

$$c_{-1} = 0, \qquad \delta_j + c_{j-1} = \epsilon'_j + 2c_j, \qquad \epsilon'_j \in \{0, 1\}. \tag{50}$$

Then

$$\Phi(\Psi(a)) - a = \sum_{j\ge0} c_j 3^j. \tag{51}$$

In particular, if the leading ternary digit of a at position L is 2, then

$$\Phi(\Psi(a)) - a \ge 3^L. \tag{52}$$

Proof. Equation equation (50) gives $\epsilon'_j - \delta_j = c_{j-1} - 2c_j$. Therefore

$$\sum_j (\epsilon'_j - \delta_j)3^j = 3\sum_j c_j 3^j - 2\sum_j c_j 3^j = \sum_j c_j 3^j.$$

If $\delta_L = 2$, then $c_L = 1$. □

This formula upgrades the inequality $K_k \ge 0$ to a finite two-state transducer: K_k is literally the base-3 evaluation of its carry mask.

6 Most-significant prefix separation

This section contains the central new structural theorem.

6.1 Aligned leading positions

Let

$$L_k = \lfloor k\beta \rfloor, \qquad u_k = \{k\beta\} \in (0, 1). \tag{53}$$

The fractional part is never zero because $\beta \notin \mathbb{Q}$. Then

$$M^k = 2^{L_k} 2^{u_k}, \qquad A^k = 3^{L_k} 3^{u_k}. \tag{54}$$

Thus the most-significant binary and ternary digit positions are both L_k. This exact alignment is special to the exponent pair $(2^x, 3^x)$.

For $0 < u < 1$, put

$$X(u) = 2^u, \qquad Y(u) = 3^u = X(u)^\vartheta. \tag{55}$$

Write the fractional expansions

$$X(u) = 1 + \sum_{t\ge1} b_t(u) 2^{-t}, \qquad b_t(u) \in \{0, 1\}, \tag{56}$$

$$Y(u) = h_0(u) + \sum_{t\geq 1} a_t(u) 3^{-t}, \qquad a_t(u) \in \{0,1,2\}, \tag{57}$$

using the nonterminating ambiguity only in the canonical terminating fashion appropriate to the integer powers in equation (54). Here $h_0(u) = 1$ for $u < \log_3 2$ and $h_0(u) = 2$ otherwise.

Lemma 6.1 (Cylinder-separation inequality)**.** *For every integer $t \geq 1$,*

$$(1 + 2^{-t})^{\vartheta} > 1 + 2\,3^{-t}. \tag{58}$$

Proof. Bernoulli's inequality and $2^{-t\vartheta} = 3^{-t}$ give

$$(1 + 2^{-t})^{\vartheta} > 1 + \vartheta 2^{-t}.$$

It remains to note that

$$\vartheta 2^{-t} > 2\,3^{-t} \quad \Longleftrightarrow \quad \vartheta \left(\frac{3}{2}\right)^t > 2,$$

and the left side is increasing in t, while at $t = 1$ it is $3\vartheta/2 > 2$. □

Theorem 6.2 (Prefix-separation theorem)**.** *Fix $u \in (0,1)$ and compare the base-2 digit string of $X(u) = 2^u$ with the base-3 digit string of $Y(u) = 3^u$, starting at their units digits and moving right.*

(a) If $u \geq \log_3 2$, the first digits are 1 and 2, so they mismatch immediately.

(b) If $0 < u < \log_3 2$, their common prefix is exactly

$$1\,0\,0 \cdots 0. \tag{59}$$

At the first mismatch after this prefix, the ternary digit is strictly larger than the binary digit.

Equivalently, after the common leading 1, the two strings can never share a digit 1.

Proof. Part (a) is immediate. For part (b), suppose the strings agree through the first $t - 1$ fractional positions, all of which must initially be zeros. If the binary digit at position t were 1, then $X(u) \geq 1 + 2^{-t}$. By lemma 6.1,

$$Y(u) = X(u)^{\vartheta} > 1 + 2\,3^{-t}.$$

Given that the preceding ternary digits are zero, its digit at position t must be 2, not 1. If instead the ternary expansion becomes nonzero first, then the binary digit is 0 and the ternary digit is 1 or 2. In every case the first mismatch has ternary digit larger than binary digit. Therefore no common 1 can occur after the leading one, and all common intervening digits are zeros. □

Corollary 6.3 (Exact first-mismatch depth)**.** *For $0 < u < \log_3 2$, define*

$$t(u) = \min\{t \geq 1 : 3^u - 1 \geq 3^{-t}\} = \left\lceil \log_3 \frac{1}{3^u - 1} \right\rceil. \tag{60}$$

Then the common prefix has the leading 1 followed by exactly $t(u)-1$ zeros, and the first mismatch is at fractional position $t(u)$. For the integer powers M^k, A^k, the degree of P_k is

$$J_k = \begin{cases} L_k, & u_k \geq \log_3 2, \\ L_k - t(u_k), & 0 < u_k < \log_3 2. \end{cases} \tag{61}$$

The leading coefficient of P_k is exactly the positive digit difference at this first mismatch.

Proof. The first nonzero ternary fractional digit occurs exactly when the ternary remainder $3^u - 1$ reaches 3^{-t}. The rest follows from theorem 6.2 after scaling by 2^{L_k} and 3^{L_k}. □

Remark 6.4 (A symbolic rigidity not visible to real approximation)**.** The theorem is stronger than the convexity inequality $\Phi(M^k) < A^k$. It identifies the most-significant mismatch and proves that every aligned digit above it is forced: a single 1, then only zeros. No arbitrary common word is possible. Thus the cross-base digit comparison has a one-parameter cylinder geometry rather than a full shift-space geometry.

6.2 Separation near the top

Without any transcendence estimate, equidistribution already gives a statistical form of top separation. An effective all-k form follows from linear forms in logarithms.

Proposition 6.5 (Density-one bounded prefix depth)**.** *For every $\eta \in (0,1)$, there is an integer T_η such that the set of k for which*

$$J_k \geq L_k - T_\eta \tag{62}$$

has natural density at least $1-\eta$.

Proof. The irrationality of β implies that $u_k = \{k\beta\}$ is uniformly distributed modulo one. Outside an interval $(0,\delta)$ of measure δ, the quantity $3^{u_k} - 1$ is bounded below, so equation (60) is bounded above by a constant depending only on δ. Choose $\delta = \eta$. □

Theorem 6.6 (Effective logarithmic prefix bound)**.** *For a fixed hypothetical pair (M,A), there is an effectively computable constant $C_M > 0$ such that, for every $k \geq 2$,*

$$t(u_k) \leq C_M \log k \quad \textit{whenever } u_k < \log_3 2. \tag{63}$$

Consequently,

$$J_k \geq L_k - C_M \log k. \tag{64}$$

Proof. The nonzero linear form

$$\Lambda_k = k \log M - L_k \log 2 = u_k \log 2 \tag{65}$$

has algebraic arguments M and 2. An explicit lower bound for a linear form in two logarithms (for example, a specialization of Matveev's theorem) yields

$$u_k \geq k^{-C_0(M)} \tag{66}$$

for $k \geq 2$, after enlarging the effective constant. On $0 < u < \log_3 2$, the mean value theorem gives $3^u - 1 \geq (\log 3)u$. Substitute this into equation (60):

$$t(u_k) \leq 1 + \log_3 \frac{1}{(\log 3)u_k} \leq C_M \log k.$$

Equation equation (64) follows from equation (61). □

Corollary 6.7 (Exponential size of the direct defect)**.** *For a fixed hypothetical pair there is an effective $C > 0$ such that*

$$D_k \geq \frac{3^{J_k}+1}{2} \gg_{M,A} \frac{A^k}{k^C} \qquad (k \geq 2). \tag{67}$$

Proof. Apply lemma 5.3 and $3^{J_k} \geq 3^{L_k} k^{-C'}$. Since $A^k = 3^{L_k+u_k} < 3^{L_k+1}$, the result follows. □

7 Macroscopic defect laws

The preceding digit theorem identifies the mismatch position. Independent convexity estimates produce explicit lower envelopes for the relative defects.

7.1 Explicit envelope functions

Define, for $0 < u < 1$,

$$\mathfrak{d}(u) = \frac{3^u - 1 - (2^u - 1)^\vartheta}{3^u}. \tag{68}$$

For $y = 3^u$ and $h = \lfloor y \rfloor \in \{1, 2\}$, define

$$\mathfrak{e}(u) = \frac{h + (y-h)^\varrho - 2^u}{2^u}. \tag{69}$$

Both functions are positive on $(0, 1)$. They have positive left limits at 1, and their only vanishing endpoint is $u = 0$.

Theorem 7.1 (Relative defect envelopes)**.** *For every $k \geq 1$,*

$$\frac{D_k}{A^k} \geq \mathfrak{d}(u_k), \tag{70}$$

$$\frac{E_k}{M^k} \geq \mathfrak{e}(u_k). \tag{71}$$

Proof. Write $M^k = 2^{L_k} + r$, where $r/2^{L_k} = 2^{u_k} - 1$. By $\Phi(r) \leq r^\vartheta$,

$$\Phi(M^k) = 3^{L_k} + \Phi(r) \leq 3^{L_k}(1 + (2^{u_k} - 1)^\vartheta).$$

Subtract from $A^k = 3^{L_k} 3^{u_k}$ and divide by A^k.

For the second estimate, write $A^k = h3^{L_k} + s$, where $s/3^{L_k} = 3^{u_k} - h$. By $\Psi(s) \geq s^\varrho$,

$$\Psi(A^k) = h2^{L_k} + \Psi(s) \geq 2^{L_k}(h + (3^{u_k} - h)^\varrho).$$

Subtract $M^k = 2^{L_k} 2^{u_k}$ and divide by it. Strict positivity is the strict convexity or concavity already used in propositions 3.3 and 3.4. ☐

Corollary 7.2 (Positive-density macroscopic defects)**.** *For every $\eta \in (0, 1)$, there are constants $c_D(\eta), c_E(\eta) > 0$ such that each of the inequalities*

$$D_k \geq c_D(\eta) A^k, \qquad E_k \geq c_E(\eta) M^k \tag{72}$$

holds simultaneously on a set of k of natural density at least $1 - \eta$.

Proof. On the compact interval $[\eta, 1]$, after taking left limits at 1, both envelope functions have a positive minimum. Uniform distribution of u_k gives the asserted density. ☐

Corollary 7.3 (Positive Cesàro means)**.** *One has*

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{k \leq N} \frac{D_k}{A^k} \geq \int_0^1 \mathfrak{d}(u)\, du > 0, \tag{73}$$

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{k \leq N} \frac{E_k}{M^k} \geq \int_0^1 \mathfrak{e}(u)\, du > 0. \tag{74}$$

Proof. The envelope functions are bounded and Riemann integrable, including at the single branch point $3^u = 2$. Apply Weyl equidistribution to $u_k = \{k\beta\}$ and use theorem 7.1. ☐

7.2 Effective all-exponent lower bounds

Theorem 7.4 (Polynomial-loss macroscopic lower bounds)**.** *For a fixed hypothetical pair (M, A), there are effectively computable constants $C_1, C_2 > 0$ and $c_1, c_2 > 0$ such that, for all $k \geq 2$,*

$$D_k \geq c_1 \frac{A^k}{k^{C_1}}, \qquad E_k \geq c_2 \frac{M^k}{k^{C_2}}. \tag{75}$$

Proof. By equation (66), $u_k \geq k^{-C_0}$. Near zero, proposition 3.3 applied to the mantissa gives $\mathfrak{d}(u) \gg u$. For $\mathfrak{e}$, when u is small the leading ternary digit is 1, and equation (29) gives $\mathfrak{e}(u) \gg (3^u - 1)^{\varrho} \gg u^{\varrho}$. Away from zero, both functions have a positive lower bound. Insert the lower bound for u_k into equations (70) and (71). □

Corollary 7.5 (Only logarithmic catastrophic borrowing)**.** *Let $s_k = \lfloor \log_2 E_k \rfloor + 1$. If the borrow block in theorem 5.4 is nonempty, then its length satisfies*

$$J_k - s_k = O_{M,A}(\log k). \tag{76}$$

Proof. By theorem 7.4, $s_k \geq \log_2 M^k - O(\log k) = L_k - O(\log k)$. Since $J_k \leq L_k$, the claim follows. □

Remark 7.6 (A useful negative conclusion)**.** The defects are not candidates for bounded-gap arguments. They are exponentially large for almost all exponents and, with only a polynomial loss, for every exponent. Any successful use of D_k, E_k must exploit factorization, congruence, or digit structure rather than small absolute size.

8 Finite-place residue transfer

Proposition 8.1 (Residue transfer)**.** *For all $r \geq 1$,*

$$n \equiv m \pmod{2^r} \Longrightarrow \Phi(n) \equiv \Phi(m) \pmod{3^r}, \tag{77}$$

$$a \equiv b \pmod{3^r} \Longrightarrow \Psi(a) \equiv \Psi(b) \pmod{2^r}. \tag{78}$$

Proof. Only the first r source digits contribute modulo the r-th power of the target prime. □

The primitive normalization imported from the supplied report implies that a hypothetical pair falls into three local branches: both M and A are units at 2 and 3, respectively; M is odd while $3 \mid A$; or $2 \mid M$ while $3 \nmid A$. The branch $2 \mid M$, $3 \mid A$ is excluded by that normalization.

Proposition 8.2 (Three recurrence branches)**.** *Fix $r \geq 1$.*

(i) If M is odd and $3 \nmid A$, there are infinitely many k for which

$$D_k \equiv 0 \pmod{3^r}, \qquad E_k \equiv 0 \pmod{2^r}. \tag{79}$$

(ii) If M is odd and $3 \mid A$, there are infinitely many k for which

$$D_k \equiv -1 \pmod{3^r}, \qquad E_k \equiv -1 \pmod{2^r}. \tag{80}$$

(iii) If $2 \mid M$ and $3 \nmid A$, there are infinitely many k for which

$$D_k \equiv 1 \pmod{3^r}, \qquad E_k \equiv 1 \pmod{2^r}. \tag{81}$$

Proof. In (i), choose k divisible by the multiplicative orders of M modulo 2^r and of A modulo 3^r. Then $M^k \equiv 1 \pmod{2^r}$, $A^k \equiv 1 \pmod{3^r}$, and use proposition 8.1. In (ii), impose $M^k \equiv 1 \pmod{2^r}$ and take k large enough that $A^k \equiv 0 \pmod{3^r}$. Case (iii) is symmetric. □

Remark 8.3 (The scale obstruction)**.** For a common exponent k, lifting-the-exponent estimates on the unit side usually provide only $r = O(\log k)$. The other side may have linearly growing valuation, but the shared modulus is limited by the unit condition. Thus the forced product $2^r 3^r$ is only polynomial in k, whereas theorem 7.4 places the defects on an exponential scale. A direct product-formula contradiction misses an exponential factor.

9 Pure-radical norm gaps

We now use the imported normalization equation (4). Put

$$\alpha = \log_2 w = \log_3 \eta. \tag{82}$$

For integers $q \geq 1$, $p \in \mathbb{Z}$, define

$$g = \gcd(d, q), \qquad D = \frac{d}{g}, \qquad m = \frac{q}{g}. \tag{83}$$

Proposition 9.1 (Degree of a powered radical). *Under the exact-degree hypothesis in equation (4), η^q has degree $D = d/g$, and its minimal polynomial is*

$$X^D - \left(\frac{A}{3^a}\right)^m. \tag{84}$$

Proof. Choose integers r, s such that $rq + sd = g$. Then

$$\eta^g = (\eta^q)^r (\eta^d)^s \in \mathbb{Q}(\eta^q),$$

while $\eta^q = (\eta^g)^{q/g}$, so $\mathbb{Q}(\eta^q) = \mathbb{Q}(\eta^g)$. The exact pure-radical degree, equivalently the standard irreducibility criterion for $X^d - A/3^a$, passes to the divisor D; hence $[\mathbb{Q}(\eta^g) : \mathbb{Q}] = D$. The displayed polynomial has degree D and vanishes at η^q, so it is minimal. □

Corollary 9.2 (Integral norm numerator). *For every integer $q \geq 1$ and every integer p with $am + pD \geq 0$, the cleared norm of $\eta^q - 3^p$ is the nonzero integer*

$$R(q, p) = \left|A^m - 3^{am+pD}\right| = 3^{am+pD}\left|3^{D(q\alpha - p)} - 1\right|. \tag{85}$$

Proof. Take the norm using equation (84) and multiply by 3^{am}. The second equality follows from $A^m = 3^{am+Dq\alpha}$. □

Let p_n/q_n be consecutive continued-fraction convergents to α, and define g_n, D_n, m_n by equation (83). The exponent vector in equation (85) is

$$\mathbf{v}_n = (m_n,\ am_n + p_n D_n). \tag{86}$$

Proposition 9.3 (Bounded determinant radical ladder). *For adjacent convergents,*

$$|\det(\mathbf{v}_n, \mathbf{v}_{n+1})| = \frac{d}{g_n g_{n+1}}, \tag{87}$$

which is a positive divisor of d.

Proof. The determinant equals

$$\frac{d}{g_n g_{n+1}} |q_n p_{n+1} - q_{n+1} p_n|.$$

The final factor is 1. Since adjacent denominators are coprime, g_n and g_{n+1} are coprime divisors of d, so their product divides d. □

The determinant-gcd lemma imported from the supplied report therefore gives a fixed finite bound for adjacent gcds after removing primes dividing $3A$. In the branch $a > 0$, primitive normalization gives $3 \nmid A$, and the full adjacent gcd is bounded by a fixed integer. In every branch, an S-unit finiteness theorem implies that the union of prime divisors of $R(q_n, p_n)$ cannot remain finite. This proves unbounded prime support along the ladder, but not a new prime at every step.

9.1 A rational secant to ϑ

Set $r_n = am_n + p_n D_n$ and $B_n = |w^{q_n} - 2^{p_n}|$. Whenever $B_n \neq 0$, define

$$Q_n = \frac{2^{p_n} R(q_n, p_n)}{D_n 3^{r_n} B_n}. \tag{88}$$

If $\epsilon_n = q_n\alpha - p_n$, cancellation gives the exact identity

$$Q_n = \frac{|3^{D_n\epsilon_n} - 1|}{D_n|2^{\epsilon_n} - 1|} \longrightarrow \vartheta. \tag{89}$$

This produces rational approximants to ϑ from two coupled integer gaps. Their heights, however, are exponential in q_n, while the error is only of continued-fraction order. Thus the construction does not contradict known irrationality measures; it is recorded because the same cancellation becomes more structural in the next section.

10 The synchronized determinant-one gap ladder

The previous ladder has determinant dividing d. A better choice of the approximated logarithm makes the determinant exactly one and synchronizes the two bases with the *same* logarithmic error.

Set

$$\gamma = d \log_2 w. \tag{90}$$

By equation (4),

$$\beta = \log_2 M = \log_3 A = a + \gamma. \tag{91}$$

Let p_n/q_n be the ordinary continued-fraction convergents to γ, and put

$$\epsilon_n = q_n\gamma - p_n, \qquad r_n = aq_n + p_n. \tag{92}$$

Define the simultaneous gaps

$$B_n = \left|w^{dq_n} - 2^{p_n}\right|, \qquad R_n = |A^{q_n} - 3^{r_n}|. \tag{93}$$

Theorem 10.1 (Same-error synchronization)**.** *The gaps satisfy*

$$B_n = 2^{p_n} |2^{\epsilon_n} - 1|, \tag{94}$$
$$R_n = 3^{r_n} |3^{\epsilon_n} - 1|. \tag{95}$$

The adjacent exponent vectors (q_n, r_n) have determinant one:

$$|q_n r_{n+1} - q_{n+1} r_n| = 1. \tag{96}$$

Proof. The first pair of identities follows from $w^{dq_n} = 2^{q_n\gamma} = 2^{p_n+\epsilon_n}$ and $A^{q_n} = 3^{q_n(a+\gamma)} = 3^{r_n+\epsilon_n}$. For the determinant,

$$q_n r_{n+1} - q_{n+1} r_n = q_n p_{n+1} - q_{n+1} p_n,$$

because the terms containing a cancel. Adjacent continued-fraction convergents have determinant ± 1. □

Corollary 10.2 (Sharp adjacent coprimality)**.** *Let*

$$G_n = M^{q_n} - 2^{r_n} = \pm 2^{aq_n} B_n.$$

The odd parts of G_n and G_{n+1} are coprime. The parts of R_n, R_{n+1} away from 3 have gcd at most 2. If $3 \nmid A$, then

$$\gcd(R_n, R_{n+1}) \leq 2. \tag{97}$$

Proof. Apply the determinant-gcd lemma for near-power gaps with determinant one. For bases $(M, 2)$, the prime-to-$2M$ gcd divides both $M - 1$ and $2 - 1 = 1$; primes dividing the odd part of M cannot divide the gap, so the odd parts are coprime. For bases $(A, 3)$, the prime-to-$3A$ gcd divides $\gcd(A - 1, 3 - 1) \leq 2$; primes dividing A but not 3 cannot divide $A^{q_n} - 3^{r_n}$. If $3 \nmid A$, no 3-adic factor remains. □

10.1 The exact gap-descent square

The errors ϵ_n alternate in sign. Restrict to the infinite subsequence $\epsilon_n > 0$. For all sufficiently large indices on this subsequence,

$$0 < B_n < 2^{p_n}, \qquad 0 < R_n < 3^{r_n}. \tag{98}$$

Then

$$M^{q_n} = 2^{r_n} + 2^{aq_n} B_n, \tag{99}$$
$$A^{q_n} = 3^{r_n} + R_n. \tag{100}$$

The two summands in each line occupy disjoint digit blocks.

Theorem 10.3 (Gap-descent square)**.** *For all sufficiently large n with $\epsilon_n > 0$,*

$$\Phi(M^{q_n}) = 3^{r_n} + 3^{aq_n} \Phi(B_n), \tag{101}$$
$$\Psi(A^{q_n}) = 2^{r_n} + \Psi(R_n), \tag{102}$$

and therefore

$$D_{q_n} = R_n - 3^{aq_n} \Phi(B_n) > 0, \tag{103}$$
$$E_{q_n} = \Psi(R_n) - 2^{aq_n} B_n > 0. \tag{104}$$

Proof. Because $B_n < 2^{p_n}$, the shifted integer $2^{aq_n} B_n$ uses binary positions strictly below $r_n = aq_n + p_n$. Hence no carry occurs in equation (99), and digit transfer gives equation (101). Similarly, $R_n < 3^{r_n}$ gives equation (102). Subtract from equation (100) and equation (99); positivity is proposition 4.1. □

The theorem is an exact commutative-square phenomenon: the same convergent error produces gaps B_n, R_n; the transfer maps then compare the large-power defects to transferred versions of the small gaps.

10.2 Why the square is not contractive

Theorem 10.4 (Scale-mismatch no-go theorem)**.** *Along the positive-error subsequence,*

$$\frac{3^{aq_n} \Phi(B_n)}{R_n} = O_{M,A}(\epsilon_n^{\vartheta - 1}) \longrightarrow 0, \tag{105}$$
$$\frac{2^{aq_n} B_n}{\Psi(R_n)} = O_{M,A}(\epsilon_n^{1-\varrho}) \longrightarrow 0. \tag{106}$$

Consequently,

$$D_{q_n} \sim R_n, \qquad E_{q_n} \sim \Psi(R_n). \tag{107}$$

Both asymptotics are in the ratio sense: each quotient of the left side by the right side tends to 1.

Proof. For small positive ϵ, $2^\epsilon - 1 \asymp \epsilon$ and $3^\epsilon - 1 \asymp \epsilon$. Thus

$$B_n \asymp 2^{p_n}\epsilon_n, \qquad R_n \asymp 3^{r_n}\epsilon_n.$$

By $\Phi(B_n) \le B_n^\vartheta$,

$$3^{aq_n}\Phi(B_n) \ll 3^{aq_n}(2^{p_n}\epsilon_n)^\vartheta = 3^{r_n}\epsilon_n^\vartheta,$$

which proves equation (105). By $\Psi(R_n) \ge R_n^\varrho$,

$$\Psi(R_n) \gg (3^{r_n}\epsilon_n)^\varrho = 2^{r_n}\epsilon_n^\varrho,$$

whereas $2^{aq_n}B_n \asymp 2^{r_n}\epsilon_n$. This proves equation (106). Insert the limits into equations (103) and (104). □

Warning 10.5. The phrase "gap descent" does not mean a descent in size. The exact identities are structurally strong, but the transferred correction is asymptotically negligible relative to the target gap. Any proof that silently assumes comparable scales reverses the actual asymptotics.

11 What the new framework rules out

The results above close several tempting but invalid lines of attack.

1. **Bounded or subexponential defects.** By theorem 7.4, both D_k and E_k are exponential up to a polynomial factor. A contradiction cannot come from showing merely that they are nonzero integers.
2. **Long common arbitrary prefixes.** By theorem 6.2, the only common most-significant word is 1 followed by zeros. Any heuristic based on random agreement of binary and ternary words is modeling the wrong symbolic system.
3. **Linear-length top agreement.** By theorem 6.6, the common zero run has length only $O(\log k)$ for a fixed hypothetical pair.
4. **Uncontrolled catastrophic borrow.** A very small value of $P_k(2)$ forces a unique block of digit pairs $(0, 1)$, and corollary 7.5 limits such a block to logarithmic length once the all-k defect lower bound is used.
5. **A naive local product formula.** The strongest immediately synchronized residue powers grow only polynomially in k, while the defects are exponentially large.
6. **A contracting continued-fraction descent.** The exact square in theorem 10.3 has the opposite scale ordering; theorem 10.4 quantifies the failure.
7. **A finite global prime set for the radical gaps.** S-unit finiteness forces unbounded prime support, though it does not by itself force a contradiction with the hypothetical pair.

These eliminations are useful progress: they replace several qualitative hopes by exact asymptotic obstructions and identify where genuinely new arithmetic information must enter.

12 Two precise closing targets

The framework suggests two non-equivalent routes to a proof. Each is formulated so that it can be attacked independently and tested computationally.

12.1 Target A: prime transfer across the gap-descent square

The determinant-one ladder gives adjacent coprimality, while theorem 10.3 links the gap R_n to $\Phi(B_n)$ and $\Psi(R_n)$ to B_n. What is missing is a mechanism that forces a substantial prime factor to survive digit transfer.

Problem 12.1 (Prime-transfer problem)**.** *Let B, R be synchronized by*

$$B = 2^p(2^\epsilon - 1), \qquad R = 3^r(3^\epsilon - 1),$$

with integral B, R, $0 < \epsilon \ll 1$, and with the exponent vectors belonging to a determinant-one continued-fraction ladder. Prove that one of the following must occur infinitely often:

- *(a) a prime outside a fixed finite set divides both R and a controlled arithmetic function of $\Phi(B)$;*
- *(b) a prime outside a fixed finite set divides both B and a controlled arithmetic function of $\Psi(R)$;*
- *(c) the carry masks of $\Phi(B)$ or $\Psi(R)$ have complexity large enough to contradict the two-term identities equations (103) and (104).*

A sufficiently uniform result of this kind would clash with adjacent coprimality in corollary 10.2. Merely proving that the prime support of R_n is unbounded is not enough; the transfer must interact with neighboring indices or with the fixed bases.

12.2 Target B: endpoint valuation excess on special intervals

Equation equation (33) may be rewritten as

$$\sum_{C_k < n \le A^k} \left(2^{v_3(n)} - 1\right) = D_k - E_k. \tag{108}$$

The interval length D_k and endpoint $C_k = \Phi(M^k)$ are not generic. Likewise, equation (34) is a special 2-adic valuation sum.

Problem 12.2 (Endpoint-rigidity problem)**.** *Prove an endpoint-sensitive estimate for one of the sums*

$$\sum_{\Phi(M^k) < n \le A^k} 2^{v_3(n)}, \qquad \sum_{M^k < n \le \Psi(A^k)} 3^{v_2(n)},$$

that uses the prefix-separation pattern of theorem 6.2 and is stronger by an exponential factor than estimates based only on interval length.

The need for an exponential improvement is forced by the scale comparison in section 8. Average-order valuation estimates alone cannot close this target.

12.3 A falsifiable symbolic criterion

Criterion 12.3 (Carry-word criterion)**.** *The conjecture would follow if one could prove that, for every pair of integers $M, A > 1$ with $\log_2 M = \log_3 A \notin \mathbb{Q}$, the digit-difference polynomials P_k cannot simultaneously satisfy all three properties for all k:*

- *(i) their common most-significant prefix is $1\,0^{O(\log k)}$;*
- *(ii) their leading mismatch is positive and every near-cancellation at $z = 2$ is carried by a $(0,1)$-block of length $O(\log k)$;*
- *(iii) the two exact valuation interval identities equations (33) and (34) hold with endpoints generated by the same multiplicative orbit.*

The criterion deliberately packages only proved necessary properties. It is suitable for a finite-state or S-adic attack because every unbounded feature has already been reduced to logarithmic depth.

13 Computational reconnaissance program

A numerical search cannot test a genuine counterexample without knowing one, but it can test candidate closing lemmas on near-miss pairs and on exact synchronized gaps. The following experiments are mathematically well-posed and avoid floating-point ambiguity by using integer arithmetic.

13.1 Exact map and carry tests

For integers M in a prescribed range, choose $A = \lfloor M^{\vartheta} \rceil$ only as a near-miss model. For $1 \le k \le K$, compute:

1. $\Phi(M^k), \Psi(A^k), D_k, E_k, K_k$;
2. the degree J_k, leading mismatch type, and borrow depth;
3. the carry mask c_j from equation (50);
4. the valuation sums in equations (33) and (34);
5. prime supports and adjacent gcds.

For a near-miss pair the exact positivity relations need not all hold; this is useful, because it reveals which patterns depend on exact exponent matching and which are generic digit phenomena.

13.2 Synchronized convergent tests

For a chosen primitive tuple (a, d, w), compute convergents to $\gamma = d \log_2 w$ with certified interval arithmetic. Retain those for which B_n, R_n in equation (93) are exact integers with the required sign. Then measure:

$$\frac{3^{aq_n}\Phi(B_n)}{R_n}, \qquad \frac{2^{aq_n}B_n}{\Psi(R_n)}, \qquad \gcd(R_n, R_{n+1}), \qquad \gcd(B_n, B_{n+1})_{\mathrm{odd}}. \tag{109}$$

The predicted power laws in theorem 10.4 provide direct regression checks. More importantly, one can search for stable residue or prime-sharing phenomena not explained by determinant one.

13.3 Search for a transfer invariant

For each prime $\ell \notin \{2, 3\}$, tabulate the finite maps

$$n \pmod{2^r} \mapsto \Phi(n) \pmod{\ell}, \qquad a \pmod{3^r} \mapsto \Psi(a) \pmod{\ell}$$

for the smallest r at which they stabilize on the relevant power orbit. The objective is not to assume periodicity in the wrong modulus, but to identify a finite quotient on which multiplication by M and A interacts with the digit transducers. Any invariant that couples two adjacent convergents would be a candidate input for problem 12.1.

14 Lean formalization blueprint

The new framework is unusually well suited to formalization because its core is finite-digit algebra. A practical dependency graph is as follows.

14.1 Module structure

Module	Principal content
`DigitTransfer.Basic`	Binary and ternary digit lists; definitions of Φ, Ψ; radix recurrences; finite support.
`DigitTransfer.Carry`	Carry normalization; super/subadditivity; composition; carry-mask theorem.
`DigitTransfer.Valuation`	Increment identities and summatory formulas using v_2, v_3.
`DigitTransfer.Bounds`	Convexity/concavity, sharp constants $1/2, 2$, equality cases.
`AlaogluErdos.Defects`	Definitions of C_k, D_k, E_k, K_k, positivity, interval identities, semigroup inequalities.
`AlaogluErdos.Words`	Difference polynomial, leading sign, borrow-block lemma, congruence corollary.
`AlaogluErdos.Prefix`	Mantissa alignment, cylinder-separation inequality, prefix-separation theorem, first-mismatch formula.
`AlaogluErdos.Density`	Irrational rotation and positive-density macroscopic defect bounds.
`AlaogluErdos.Congruences`	Residue transfer and the three local branches.
`AlaogluErdos.Gaps`	Continued-fraction convergents, determinant one, exact gap-descent identities.

14.2 Suggested theorem statements

The following pseudo-Lean signatures indicate the intended granularity. They are not claimed to compile verbatim.

```
def phi (n : Nat) : Nat :=
  (Nat.digits 2 n).eval 3

def psi (n : Nat) : Nat :=
  (Nat.digits 3 n).eval 2

theorem psi_phi (n : Nat) : psi (phi n) = n

theorem phi_sub_phi_pred (n : Nat) (hn : 0 < n) :
  phi n - phi (n - 1) = (3 ^ padicValNat 2 n + 1) / 2

theorem defect_interval (k : Nat) (hk : 0 < k) :
  E M A k = 2 * D M A k -
    \sum n in Ioc (phi (M^k)) (A^k), 2 ^ padicValNat 3 n

theorem prefix_separation (u : Real) (hu0 : 0 < u) (hu1 : u < 1) :
  CommonPrefix (base2Mantissa u) (base3Mantissa u) =
    [1] ++ List.replicate (prefixZeroLength u) 0
```

14.3 Formalization order

The recommended order is:

$$\texttt{Basic} \to \texttt{Carry} \to \texttt{Bounds} \to \texttt{Defects} \to \texttt{Words} \to \texttt{Prefix}.$$

This proves the entirely elementary core before importing analytic number theory. The Matveev-based $O(\log k)$ theorem should initially be formalized as a theorem parameterized by a lower-bound hypothesis for the linear form. A later module can connect that hypothesis to an available formal library for logarithmic forms.

15 Further deductions and refinements

15.1 The first mismatch controls the direct defect

Combining corollary 6.3 and lemma 5.3 gives the fully explicit estimate

$$D_k \geq \begin{cases} (3^{L_k}+1)/2, & u_k \geq \log_3 2, \\ (3^{L_k-t(u_k)}+1)/2, & 0 < u_k < \log_3 2. \end{cases} \tag{110}$$

This bound arises from digit order, not from convexity. It can be stronger than the smooth envelope when the first mismatch digit is high.

15.2 The only dangerous leading mismatch

If the leading coefficient of P_k is 2, then $E_k \geq 2^{J_k}+1$ by lemma 5.3. Therefore any substantial cancellation in $E_k = P_k(2)$ requires a leading coefficient 1, which means that the first mismatching digit pair is exactly

$$(\delta_{k,J_k}, \epsilon_{k,J_k}) \in \{(1,0),(2,1)\}. \tag{111}$$

The following digits must then begin with the inverse pattern $(0,1)$ if E_k is small relative to 2^{J_k}. Hence every difficult word has the schematic form

$$\underbrace{(1,1)(0,0)\cdots(0,0)}_{\text{forced common top}} \quad \underbrace{(1,0) \text{ or } (2,1)}_{\text{positive mismatch}} \quad \underbrace{(0,1)\cdots(0,1)}_{\text{borrow block}} \quad \cdots. \tag{112}$$

This is a much smaller symbolic language than the unrestricted product of a binary and a ternary shift.

15.3 A compactness reduction for symbolic attacks

Fix a constant C large enough for the logarithmic bounds in theorem 6.6 and corollary 7.5. At exponent k, all nontrivial most-significant cancellation is confined to two windows of total length $O(C\log k)$: the common-zero window above J_k and any borrow window below it. Outside these windows, $P_k(2)$ and $P_k(3)$ have a stable sign contribution of exponential size.

This suggests encoding only the $O(\log k)$-window as a state. The number of states is polynomial in k, not exponential in the full digit length. A successful multiplication-by-M, multiplication-by-A transition analysis would therefore need to beat polynomial state growth. This is a concrete place where an S-unit theorem, a subspace theorem with moving targets, or an entropy estimate might enter.

15.4 Why four exponentials still appears

The new results exploit integrality much more directly than a bare algebraicity argument, but they do not create an algebraic relation between $\log 2$ and $\log 3$. The synchronized ladder reduces two exponential approximations to one continued-fraction error, yet the exact digit maps are nonlinear and nonmultiplicative. A closing theorem must therefore add arithmetic rigidity of digital carries or prime factors. Without that input, the construction remains compatible with the known four-exponentials frontier described in the supplied report.

16 Conclusions

The investigation produced the following rigorous advances.

1. Two exact digit-transfer maps Φ, Ψ convert the hypothetical power relation into positive integral defects. Their carry algebra, valuation increments, p-adic extensions, and sharp archimedean bounds are fully explicit.
2. The same digit-difference polynomial evaluates to D_k at 3 and E_k at 2. Its leading coefficient is positive, and any strong cancellation at 2 forces a long, unique $(0, 1)$ borrow block.
3. The most-significant binary and ternary words cannot share an arbitrary prefix. Their common prefix is only 1 followed by zeros, and the first mismatch favors the ternary digit. This mismatch lies within $O(\log k)$ of the top for every k, effectively for a fixed hypothetical pair.
4. Both defects are macroscopic on a set of exponents of density arbitrarily close to one. An explicit logarithmic-form estimate strengthens this to exponential lower bounds with only polynomial loss for every exponent.
5. The pure-radical reduction yields norm gaps whose adjacent determinant is bounded. Optimizing the approximated logarithm produces a new ladder with determinant exactly one and sharp adjacent coprimality.
6. On one continued-fraction sign subsequence, the ladder fits into an exact gap-descent square. A rigorous asymptotic calculation shows that the square is noncontractive, preventing an illusory descent proof and exposing the missing scale.

The conjecture is therefore not proved here. The most credible next step is not another generic approximation to ϑ, but one of two highly specific inputs: a prime-transfer theorem across the determinant-one square, or an endpoint-sensitive valuation theorem for the exact defect intervals. Both targets now come with rigid digit words, logarithmic state windows, and explicit failure thresholds, making them substantially more falsifiable than the broad approaches they replace.

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